ENTERPRISE ARCHITECTURE DOSSIER

PathaSathi Global Operating System

Full System Architecture Mindmap & Multi-Level Data Flow Diagrams (DFD)

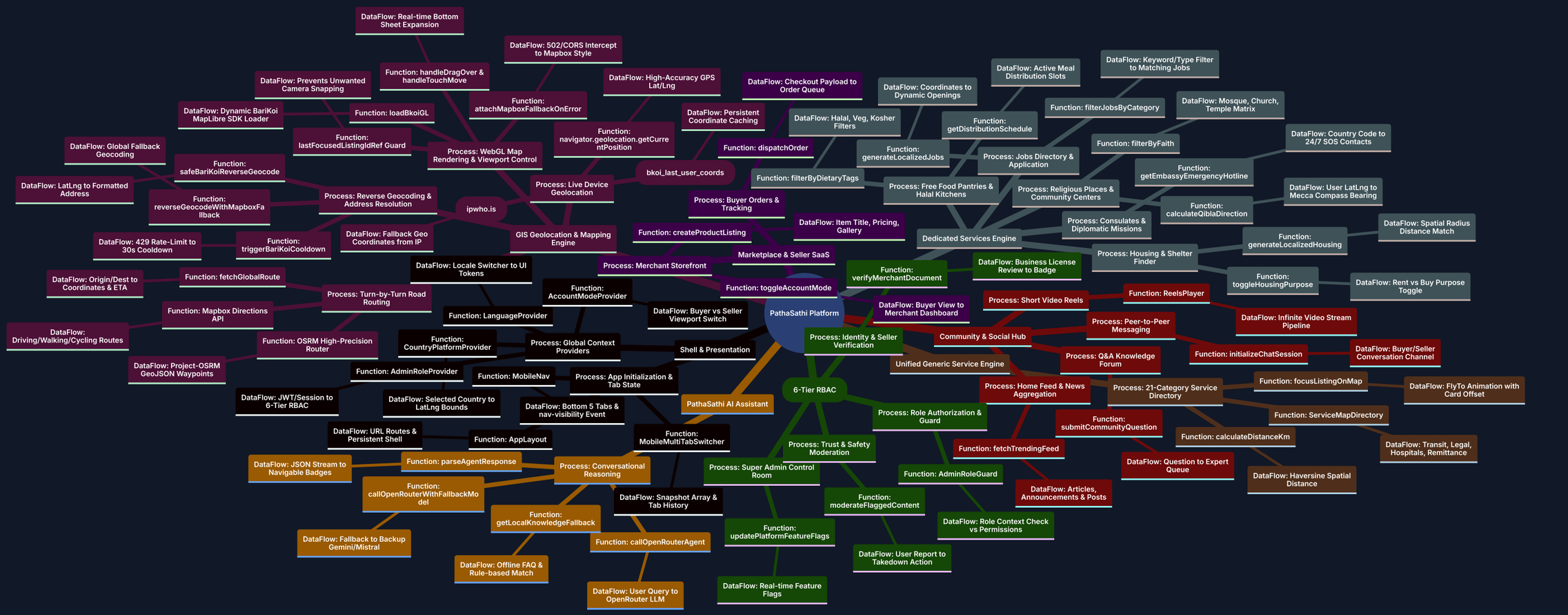
Complete architectural specifications, hardware and cloud integration pipelines, spatial GIS algorithms, turn-by-turn routing mechanics, conversational AI engines, and 6-tier governance RBAC.

Platform: PathaSathi Global **Architecture Version:** 3.5 Enterprise **Diagram Levels:** DFD Level 0 to Level 3.4

SYSTEM ARCHITECTURE OVERVIEW

Full System Architectural Mindmap

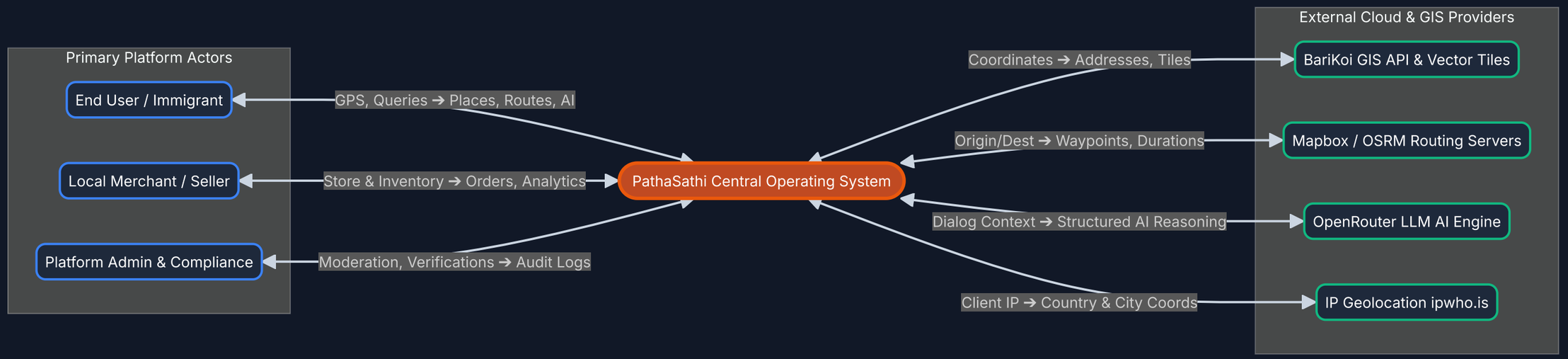
Comprehensive structural breakdown of PathaSathi platform across Presentation Shell, GIS Engine, Dedicated Services, Generic Directory, Community, Marketplace, AI Assistant, and Admin Governance.



DATA FLOW ARCHITECTURE

DFD Level 0: Context Diagram

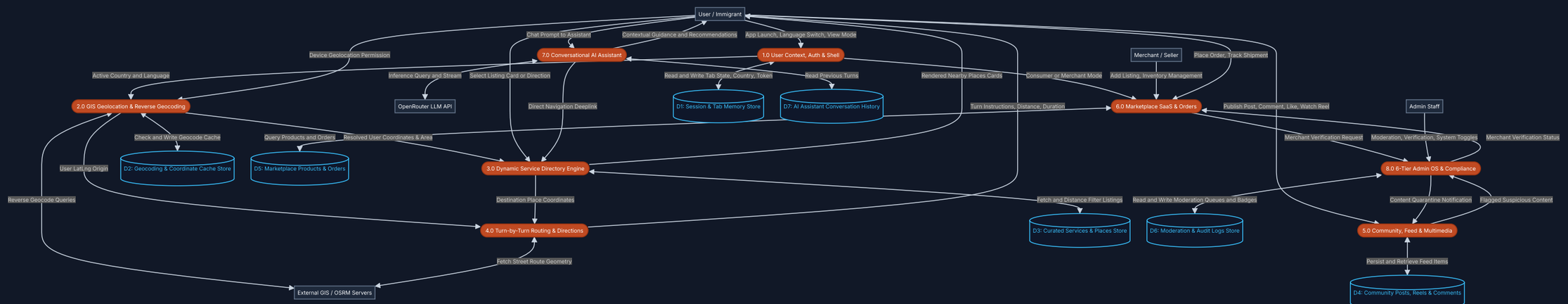
High-level data exchange boundaries connecting primary platform actors (End Users, Merchants, Admin Staff) to the central PathaSathi Operating System and third-party cloud engines.



SYSTEM SUBSYSTEMS & DATA STORES

DFD Level 1: System-Level Decomposition

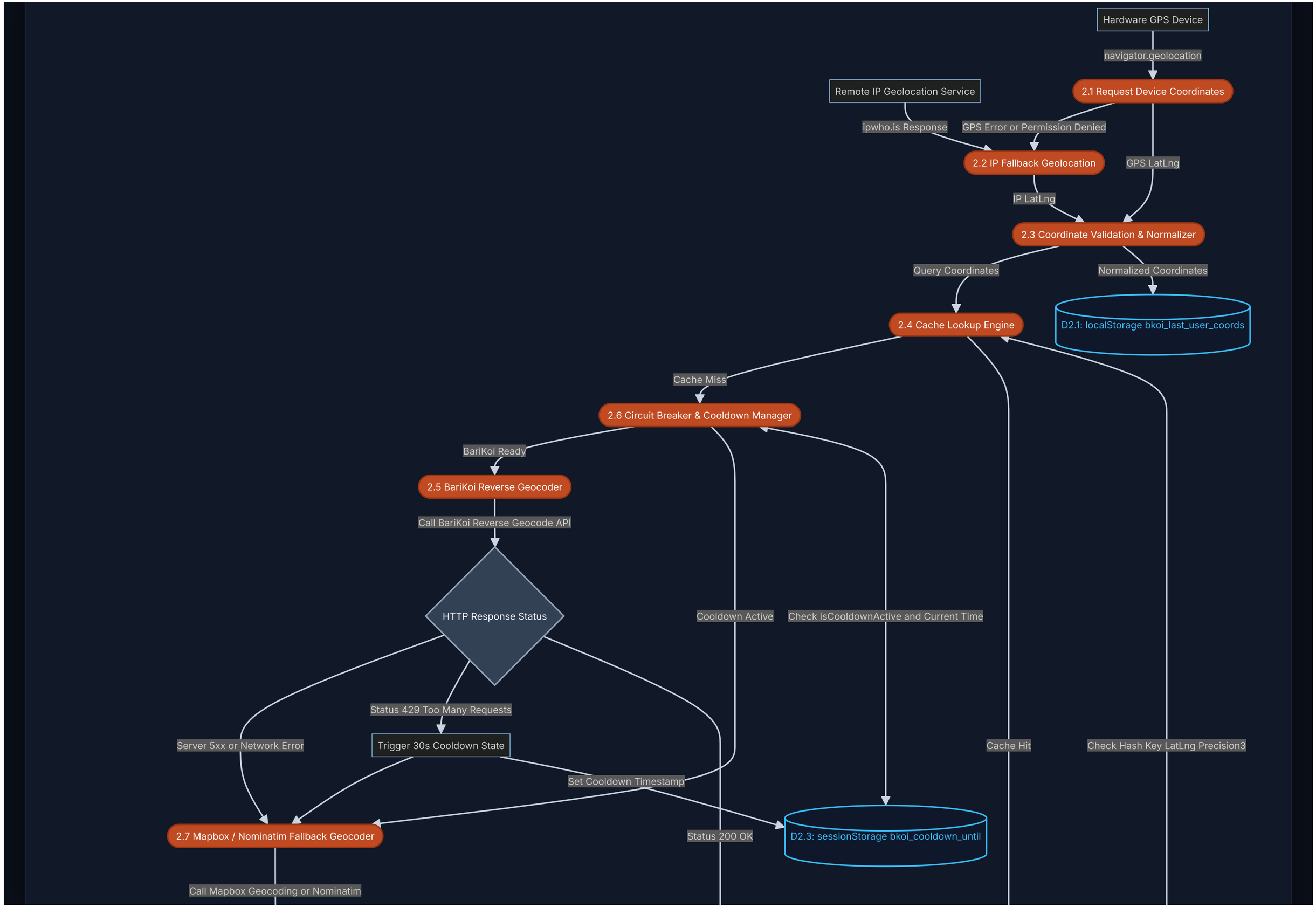
Decomposition of PathaSathi into 8 core functional subsystems with interconnected Data Stores D1 through D7.

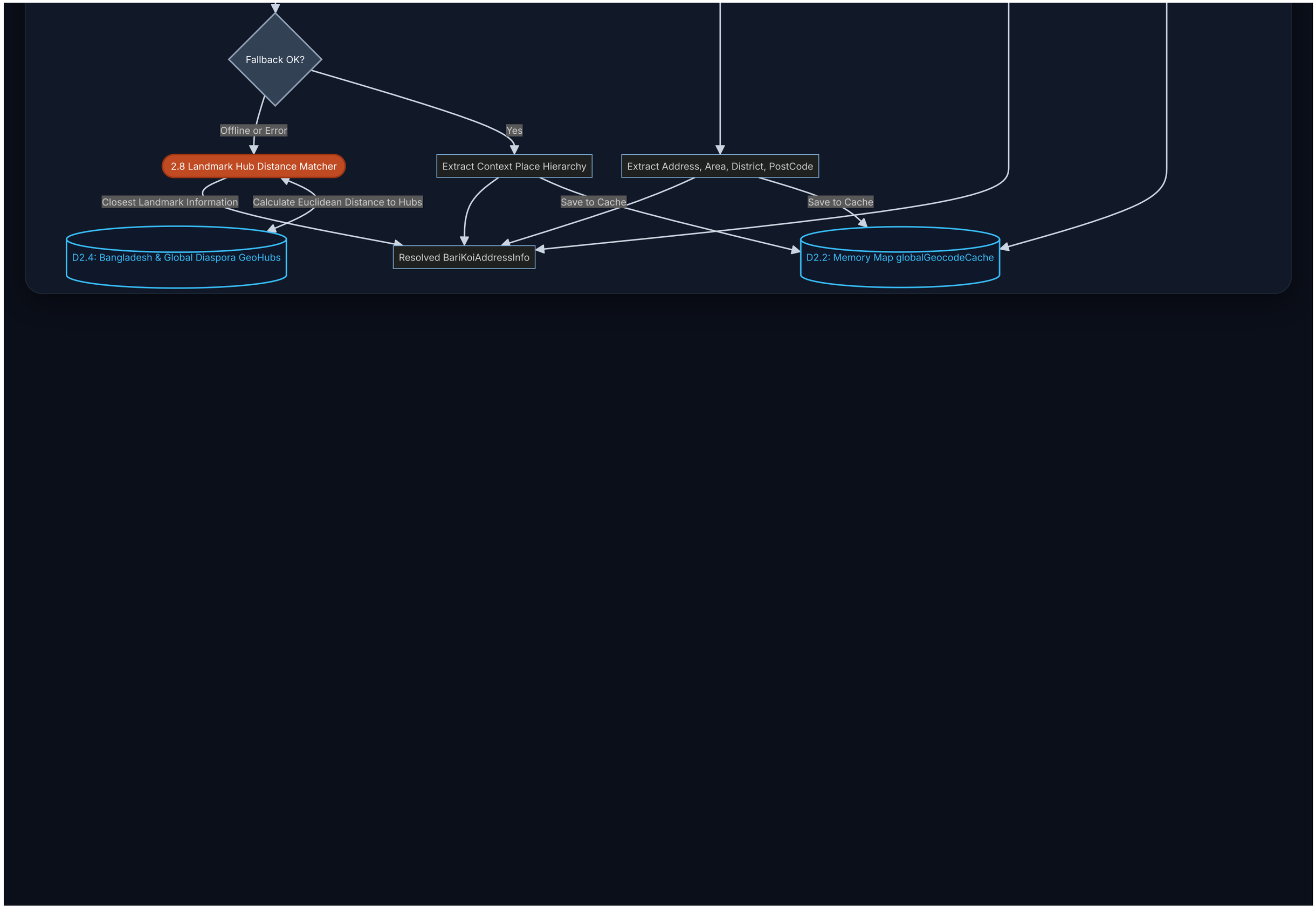


GIS & SPATIAL PIPELINE

DFD Level 2.0: GIS Geolocation & Reverse Geocoding Process

Device coordinate resolution pipeline from GPS acquisition, IP fallback, multi-tier cache check, BariKoi reverse geocoding with 429 rate limit circuit breaker, to diaspora hub matching.

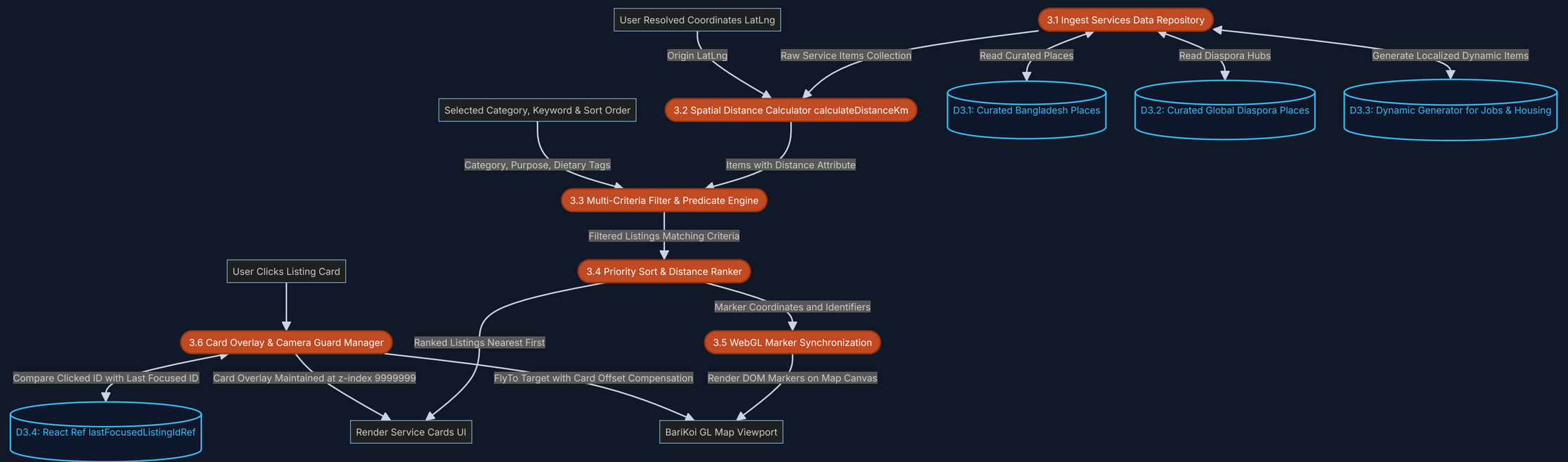

Hardware GPS Device
navigator.geolocation
Remote IP Geolocation Service
2.1 Request Device Coordinates
ipwho.is Response
GPS Error or Permission Denied
2.2 IP Fallback Geolocation
GPS LatLng
IP LatLng
2.3 Coordinate Validation & Normalizer
Query Coordinates
Normalized Coordinates
D2.1: localStorage bkoi_last_user_coords
2.4 Cache Lookup Engine
Cache Miss
2.6 Circuit Breaker & Cooldown Manager
BariKoi Ready
2.5 BariKoi Reverse Geocoder
Call BariKoi Reverse Geocode API
HTTP Response Status
Cooldown Active
Check isCooldownActive and Current Time
Status 429 Too Many Requests
Server 5xx or Network Error
Trigger 30s Cooldown State
Cache Hit
Check Hash Key LatLng Precision3
Set Cooldown Timestamp
2.7 Mapbox / Nominatim Fallback Geocoder
Status 200 OK
D2.3: sessionStorage bkoi_cooldown_until
Call Mapbox Geocoding or Nominatim


Fallback OK?
Offline or Error
Yes
2.8 Landmark Hub Distance Matcher
Extract Context Place Hierarchy
Extract Address, Area, District, PostCode
Closest Landmark Information
Calculate Euclidean Distance to Hubs
Save to Cache
Save to Cache
D2.4: Bangladesh & Global Diaspora GeoHubs
Resolved BariKoiAddressInfo
D2.2: Memory Map globalGeocodeCache

LISTING & DISCOVERY ENGINE

DFD Level 2.1: Dynamic Service Directory & Spatial Query Process

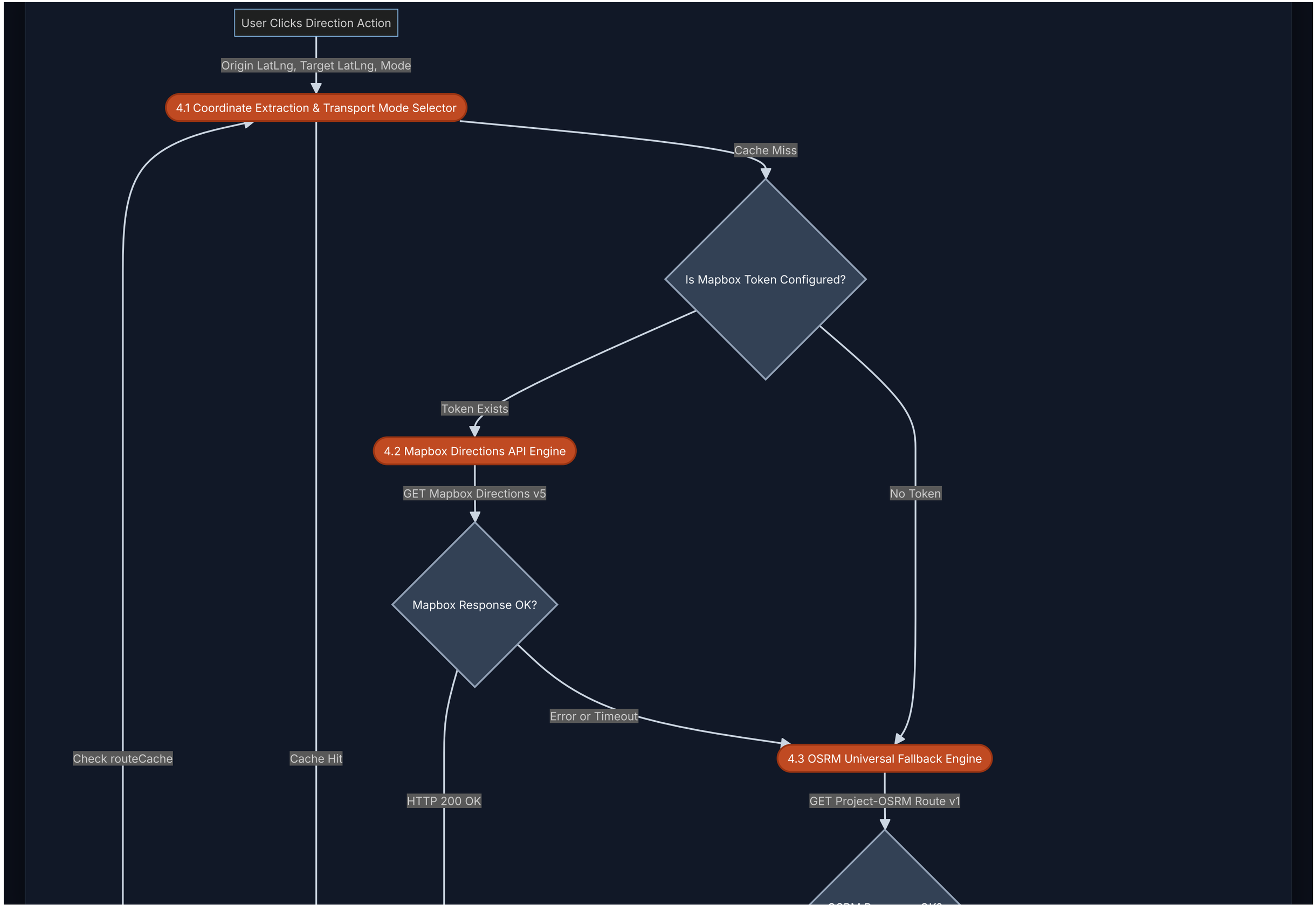
Spatial distance calculation (calculateDistanceKm), multi-criteria filtering, priority ranker, WebGL marker sync, and card overlay camera snapping guard.

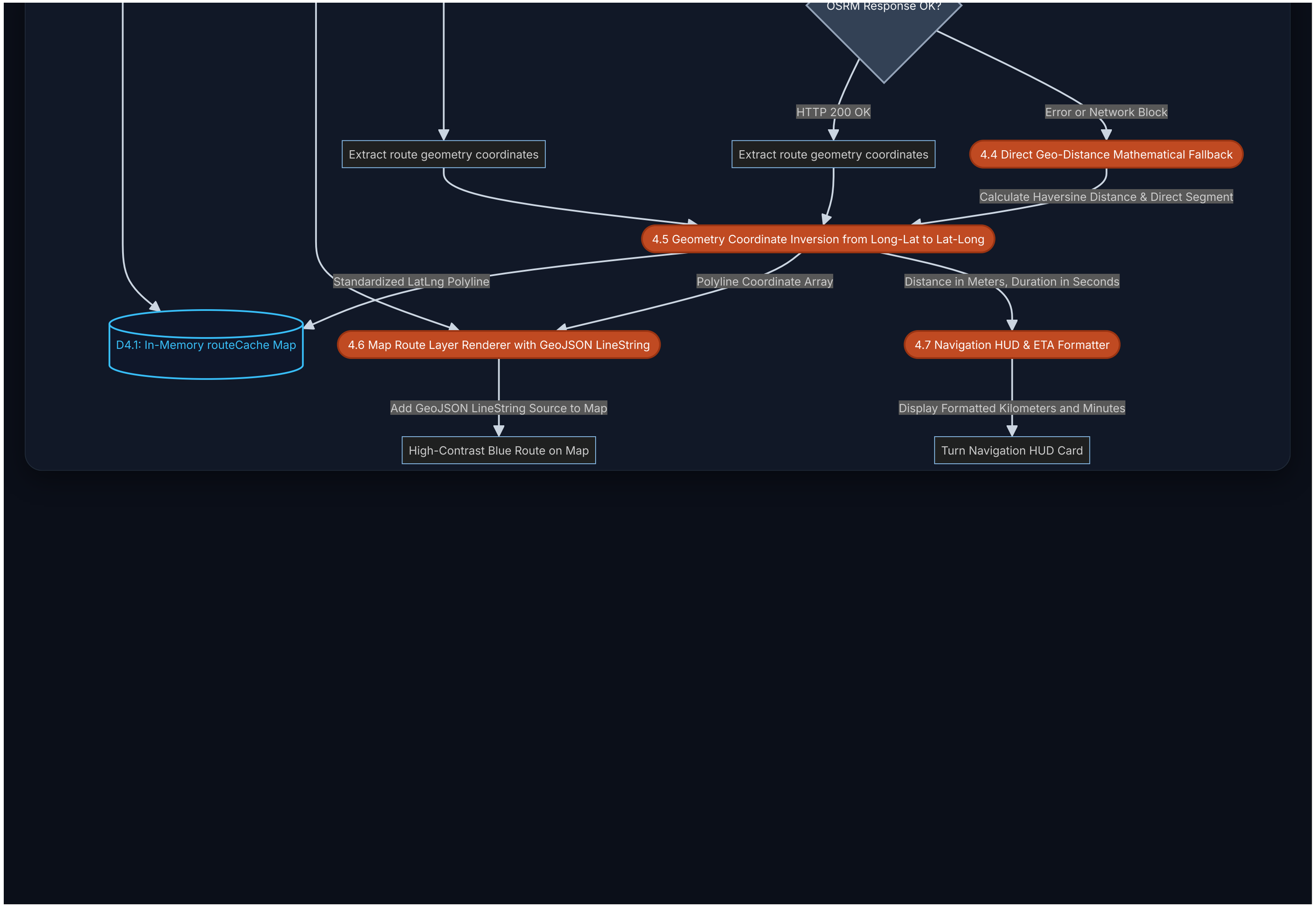


NAVIGATION & DIRECTIONS ENGINE

DFD Level 2.2: Turn-by-Turn Routing & Directions Process

Coordinate extraction, Mapbox profile selection, OSRM universal fallback, direct line distance fallback, coordinate inversion ([lng,lat]→[lat,lng]), and navigation HUD.

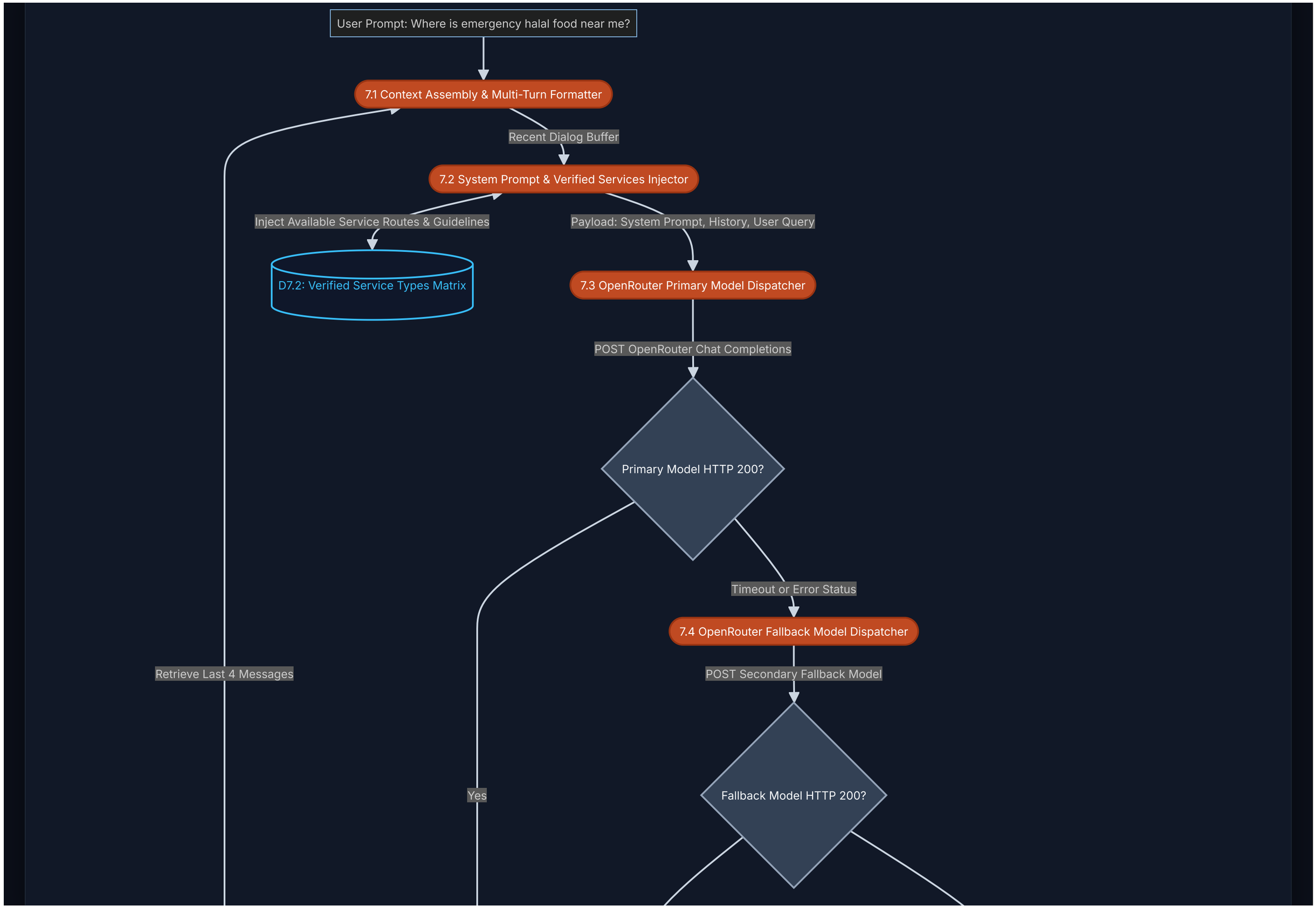

User Clicks Direction Action
Origin LatLng, Target LatLng, Mode
4.1 Coordinate Extraction & Transport Mode Selector
Check routeCache
Cache Hit
Cache Miss
Is Mapbox Token Configured?
Token Exists
No Token
4.2 Mapbox Directions API Engine
GET Mapbox Directions v5
Mapbox Response OK?
HTTP 200 OK
Error or Timeout
4.3 OSRM Universal Fallback Engine
GET Project-OSRM Route v1

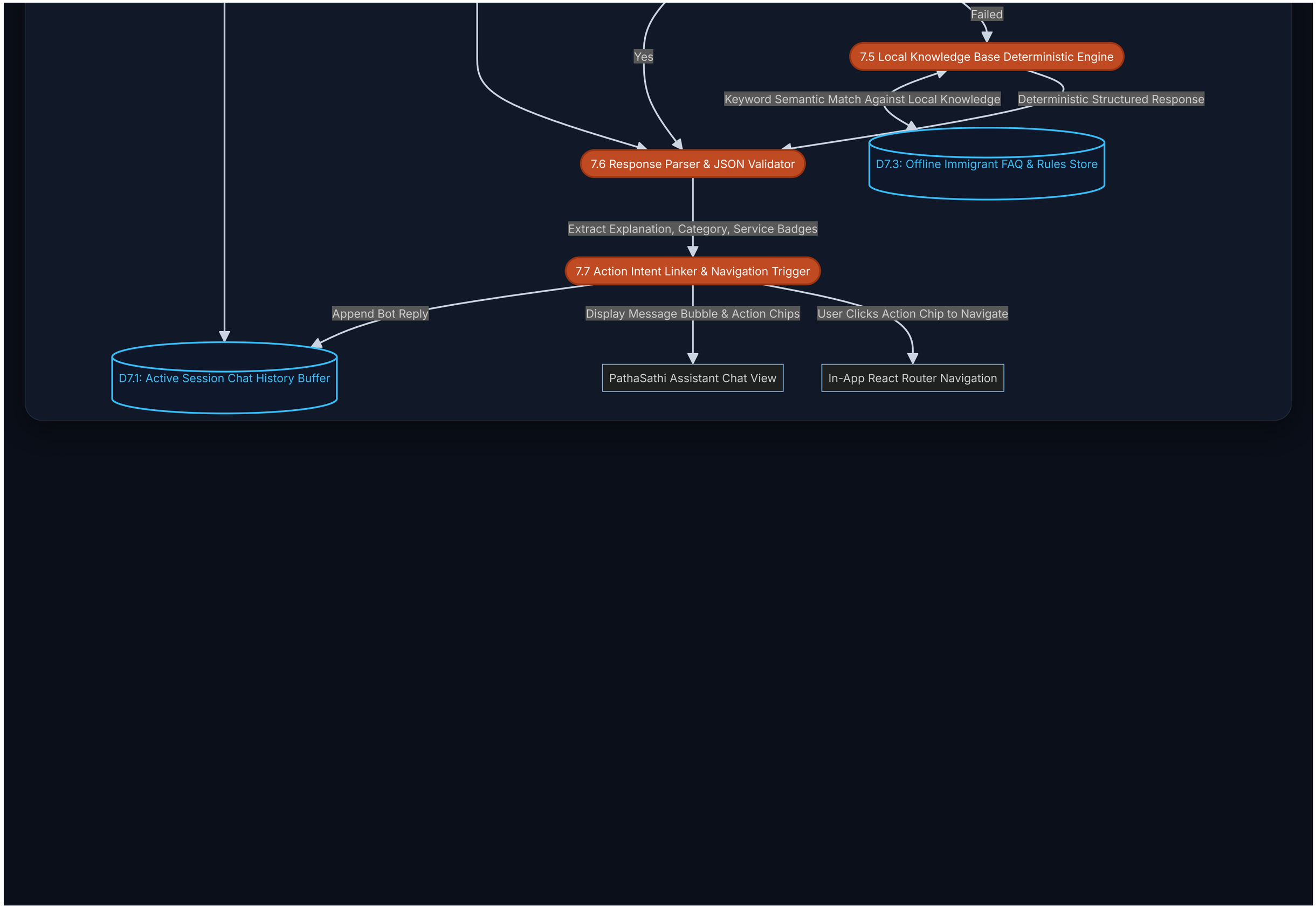

OSRM Response OK?
HTTP 200 OK
Error or Network Block
Extract route geometry coordinates
Extract route geometry coordinates
4.4 Direct Geo-Distance Mathematical Fallback
Calculate Haversine Distance & Direct Segment
4.5 Geometry Coordinate Inversion from Long-Lat to Lat-Long
Standardized LatLng Polyline
Polyline Coordinate Array
Distance in Meters, Duration in Seconds
D4.1: In-Memory routeCache Map
4.6 Map Route Layer Renderer with GeoJSON LineString
4.7 Navigation HUD & ETA Formatter
Add GeoJSON LineString Source to Map
Display Formatted Kilometers and Minutes
High-Contrast Blue Route on Map
Turn Navigation HUD Card

ARTIFICIAL INTELLIGENCE & GUIDANCE

DFD Level 2.3: Conversational AI Assistant Process

Conversational memory buffering, system prompt injection, OpenRouter primary model invocation, fallback model failover, deterministic immigrant FAQ rules, and in-app action link dispatch.

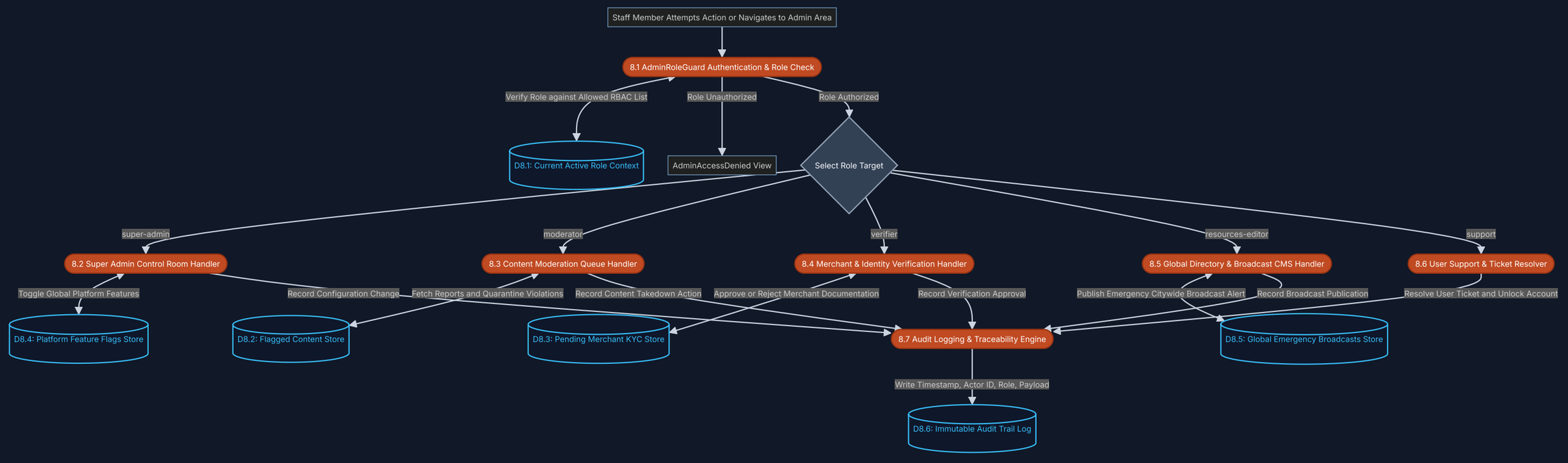

User Prompt: Where is emergency halal food near me?
7.1 Context Assembly & Multi-Turn Formatter
Recent Dialog Buffer
7.2 System Prompt & Verified Services Injector
Inject Available Service Routes & Guidelines
Payload: System Prompt, History, User Query
D7.2: Verified Service Types Matrix
7.3 OpenRouter Primary Model Dispatcher
POST OpenRouter Chat Completions
Primary Model HTTP 200?
Timeout or Error Status
7.4 OpenRouter Fallback Model Dispatcher
POST Secondary Fallback Model
Fallback Model HTTP 200?
Retrieve Last 4 Messages
Yes


Failed
Yes
7.5 Local Knowledge Base Deterministic Engine
Keyword Semantic Match Against Local Knowledge
Deterministic Structured Response
7.6 Response Parser & JSON Validator
D7.3: Offline Immigrant FAQ & Rules Store
Extract Explanation, Category, Service Badges
7.7 Action Intent Linker & Navigation Trigger
Append Bot Reply
Display Message Bubble & Action Chips
User Clicks Action Chip to Navigate
D7.1: Active Session Chat History Buffer
PathaSathi Assistant Chat View
In-App React Router Navigation

GOVERNANCE & COMPLIANCE

DFD Level 2.4: 6-Tier Admin Operating System & RBAC Process

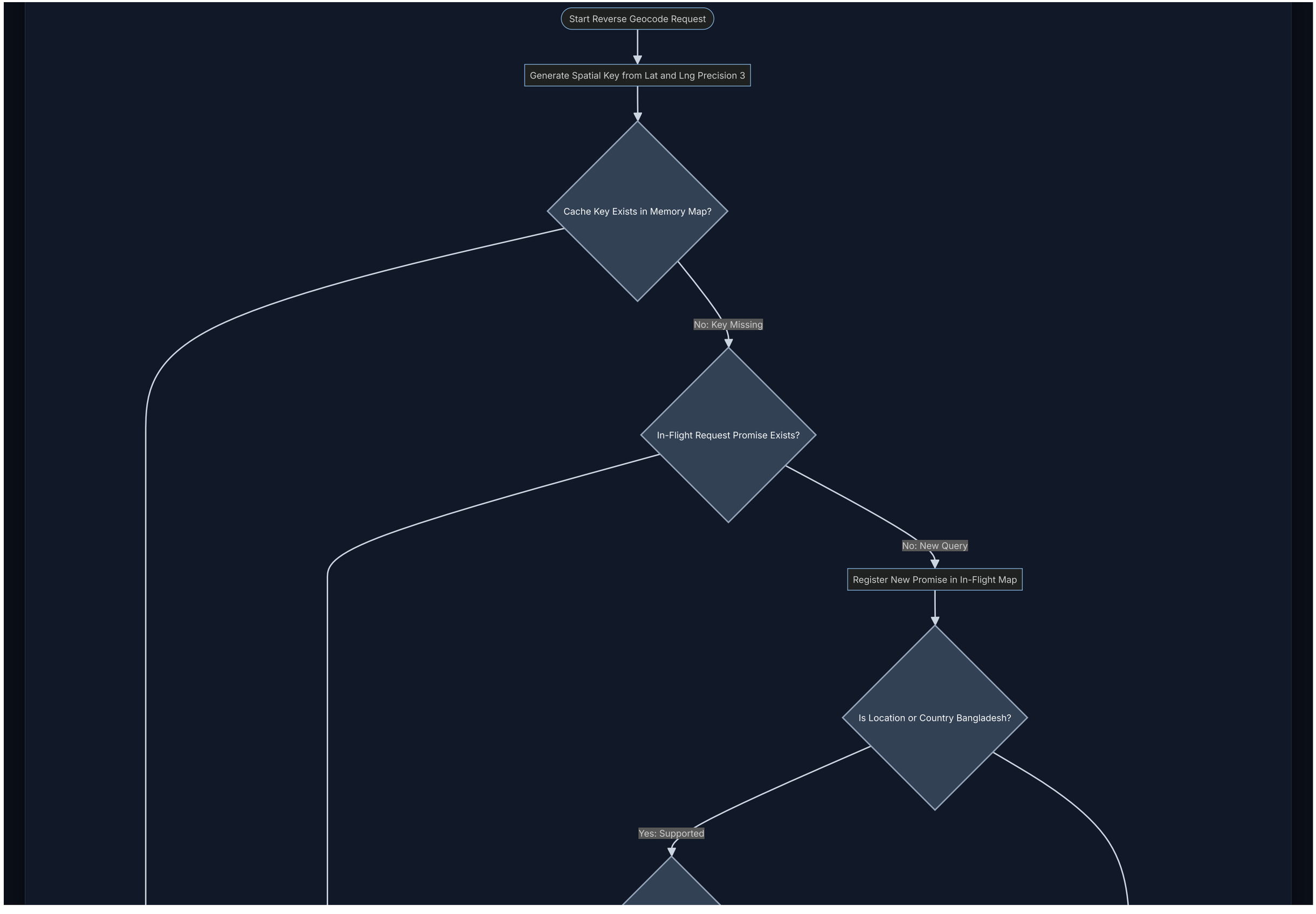
AdminRoleGuard route gatekeeper, Super Admin control room, Moderator queue, Verifier KYC review, Resource/Directory broadcast CMS, Support desk, and immutable audit logger.

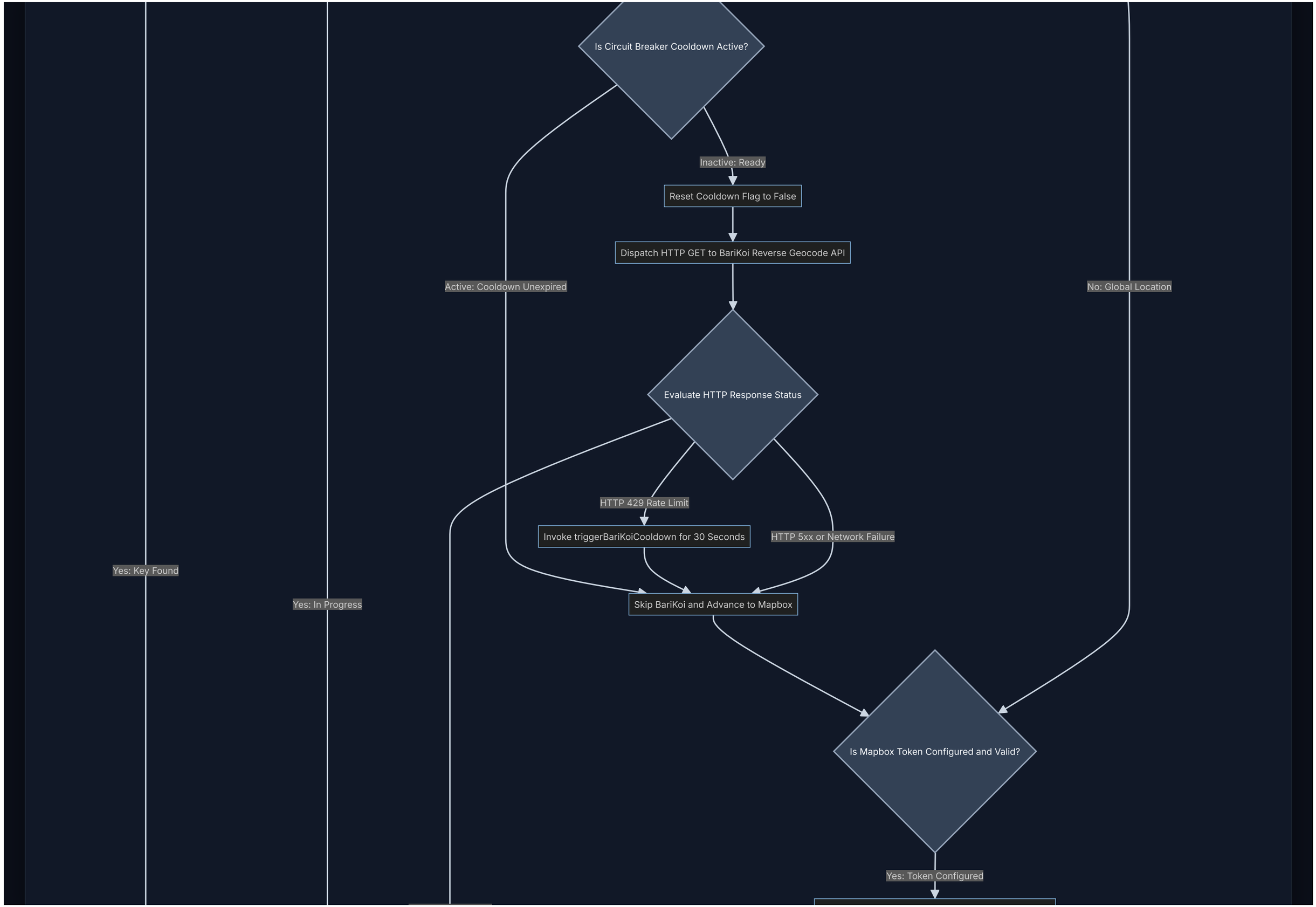


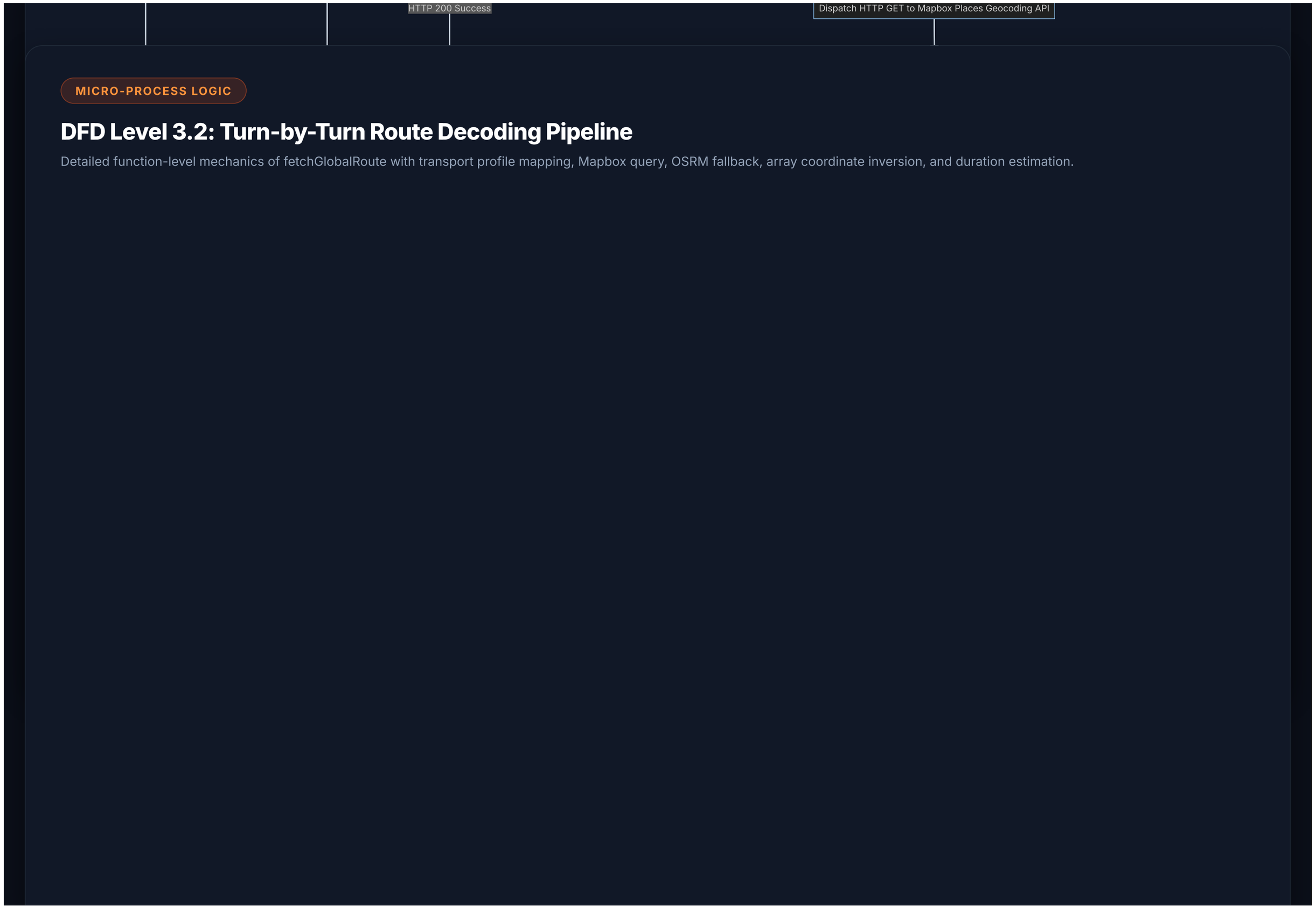
MICRO-PROCESS LOGIC

DFD Level 3.1: BariKoi Reverse Geocoding with Circuit Breaker Pipeline

Detailed function-level loop of safeBariKoiReverseGeocode with spatial precision hashing, in-flight request deduplication, 429 rate limit tripping, Mapbox fallback, and diaspora hub matching.


Start Reverse Geocode Request
Generate Spatial Key from Lat and Lng Precision 3
Cache Key Exists in Memory Map?
No: Key Missing
In-Flight Request Promise Exists?
No: New Query
Register New Promise in In-Flight Map
Is Location or Country Bangladesh?
Yes: Supported

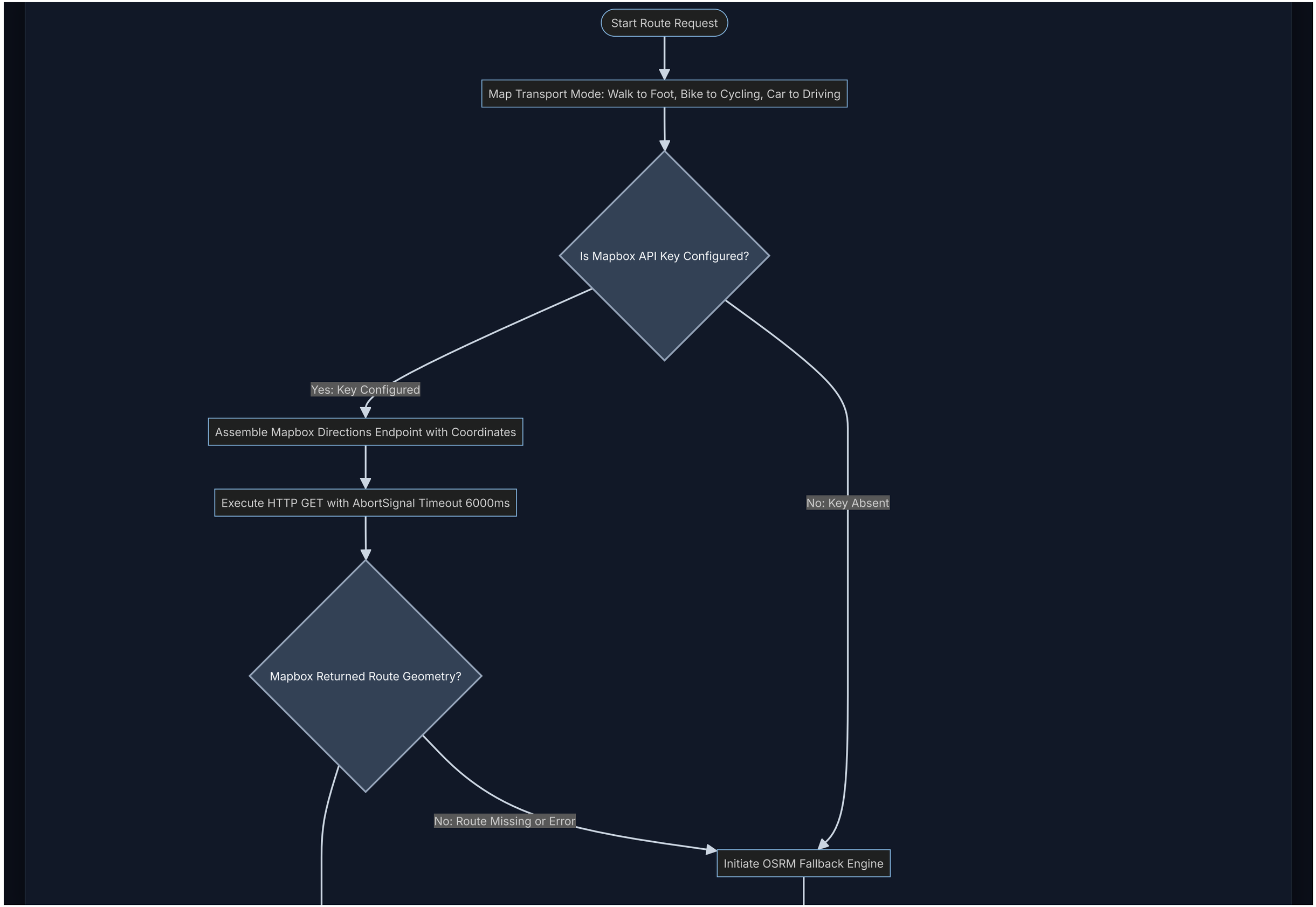

Is Circuit Breaker Cooldown Active?
Inactive: Ready
Reset Cooldown Flag to False
Dispatch HTTP GET to BariKoi Reverse Geocode API
Active: Cooldown Unexpired
No: Global Location
Evaluate HTTP Response Status
HTTP 429 Rate Limit
Invoke triggerBariKoiCooldown for 30 Seconds
HTTP 5xx or Network Failure
Yes: Key Found
Yes: In Progress
Skip BariKoi and Advance to Mapbox
Is Mapbox Token Configured and Valid?
Yes: Token Configured

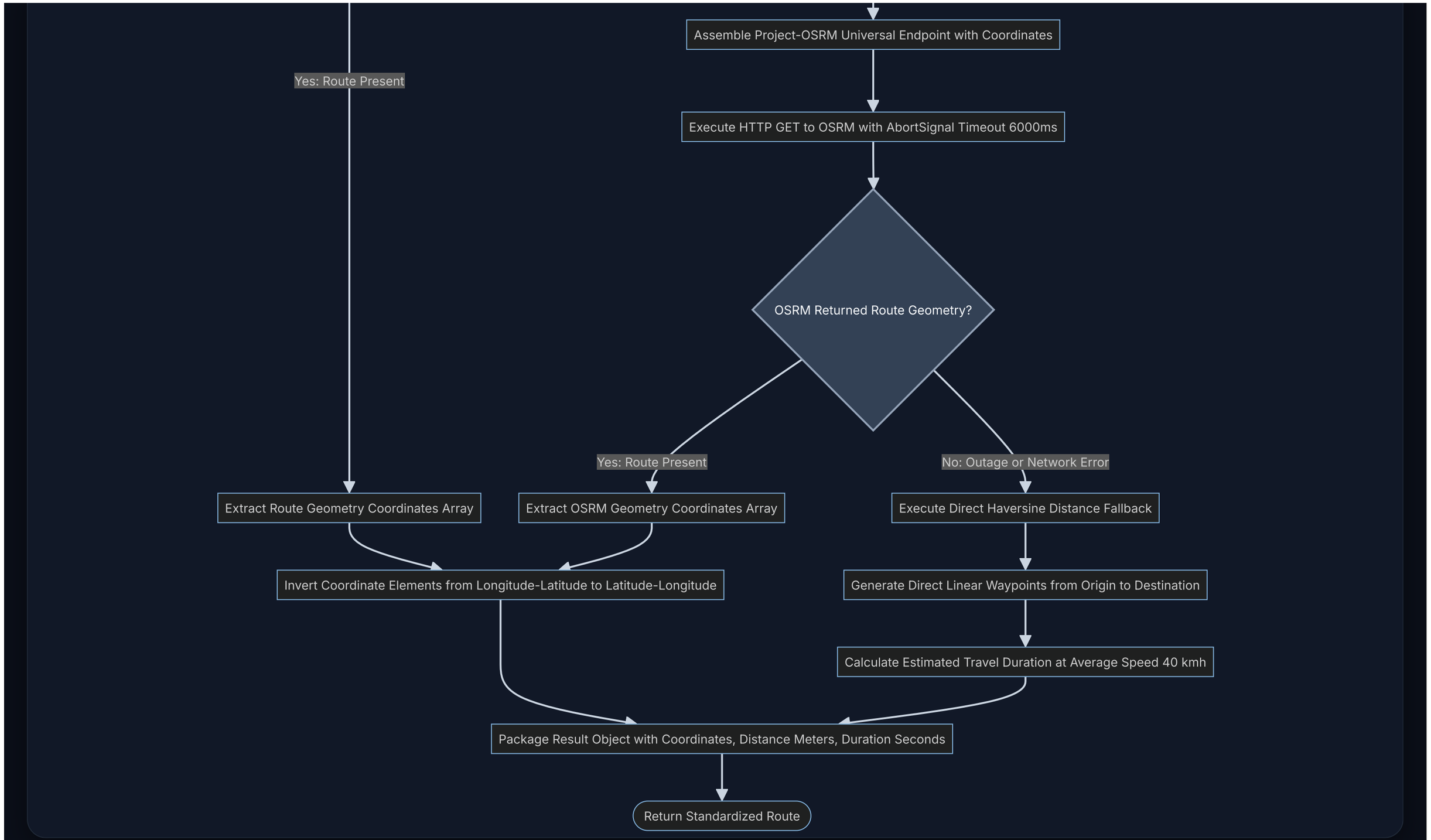


MICRO-PROCESS LOGIC

DFD Level 3.2: Turn-by-Turn Route Decoding Pipeline

Detailed function-level mechanics of fetchGlobalRoute with transport profile mapping, Mapbox query, OSRM fallback, array coordinate inversion, and duration estimation.

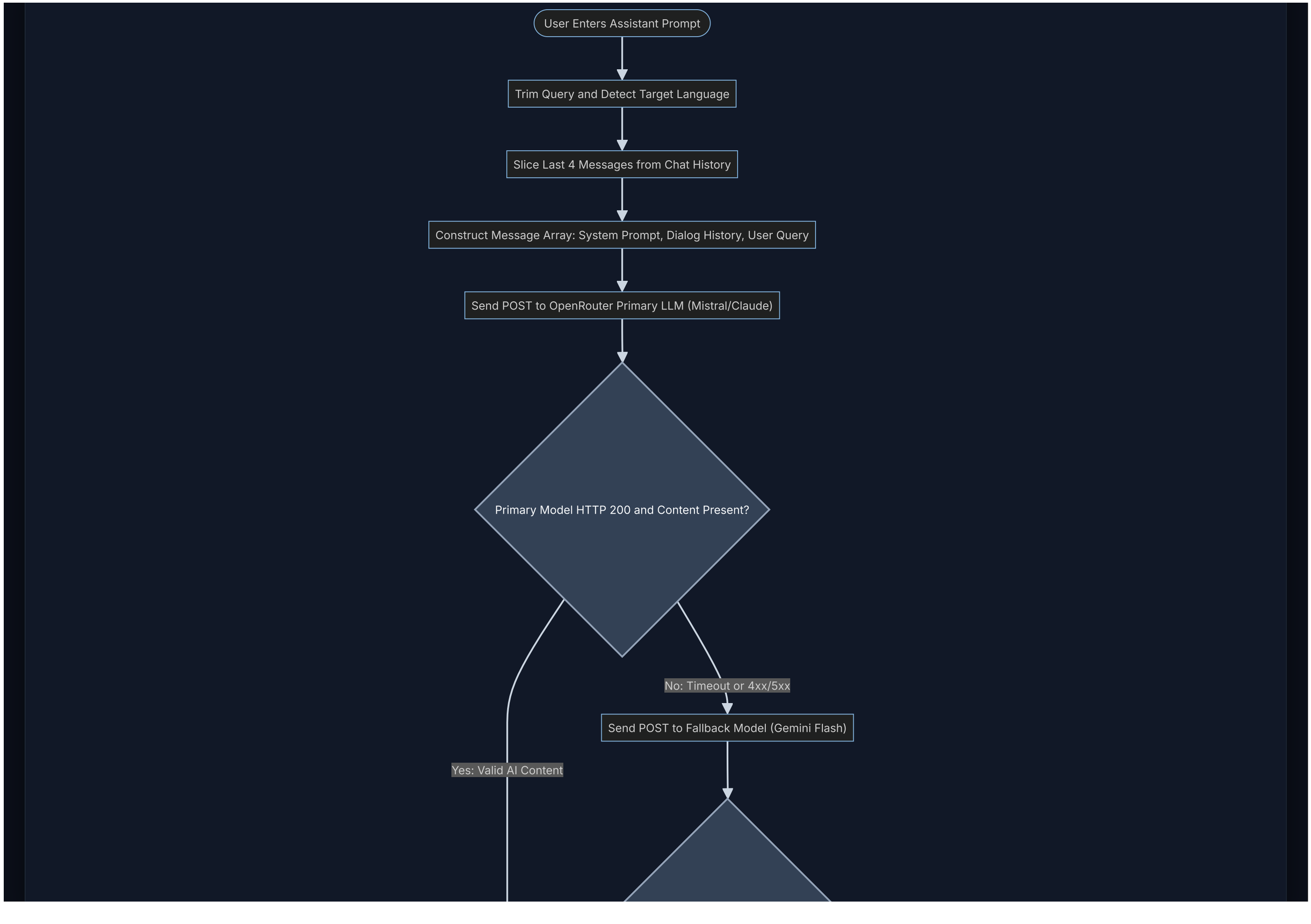

Start Route Request
Map Transport Mode: Walk to Foot, Bike to Cycling, Car to Driving
Is Mapbox API Key Configured?
Yes: Key Configured
Assemble Mapbox Directions Endpoint with Coordinates
Execute HTTP GET with AbortSignal Timeout 6000ms
Mapbox Returned Route Geometry?
No: Route Missing or Error
No: Key Absent
Initiate OSRM Fallback Engine

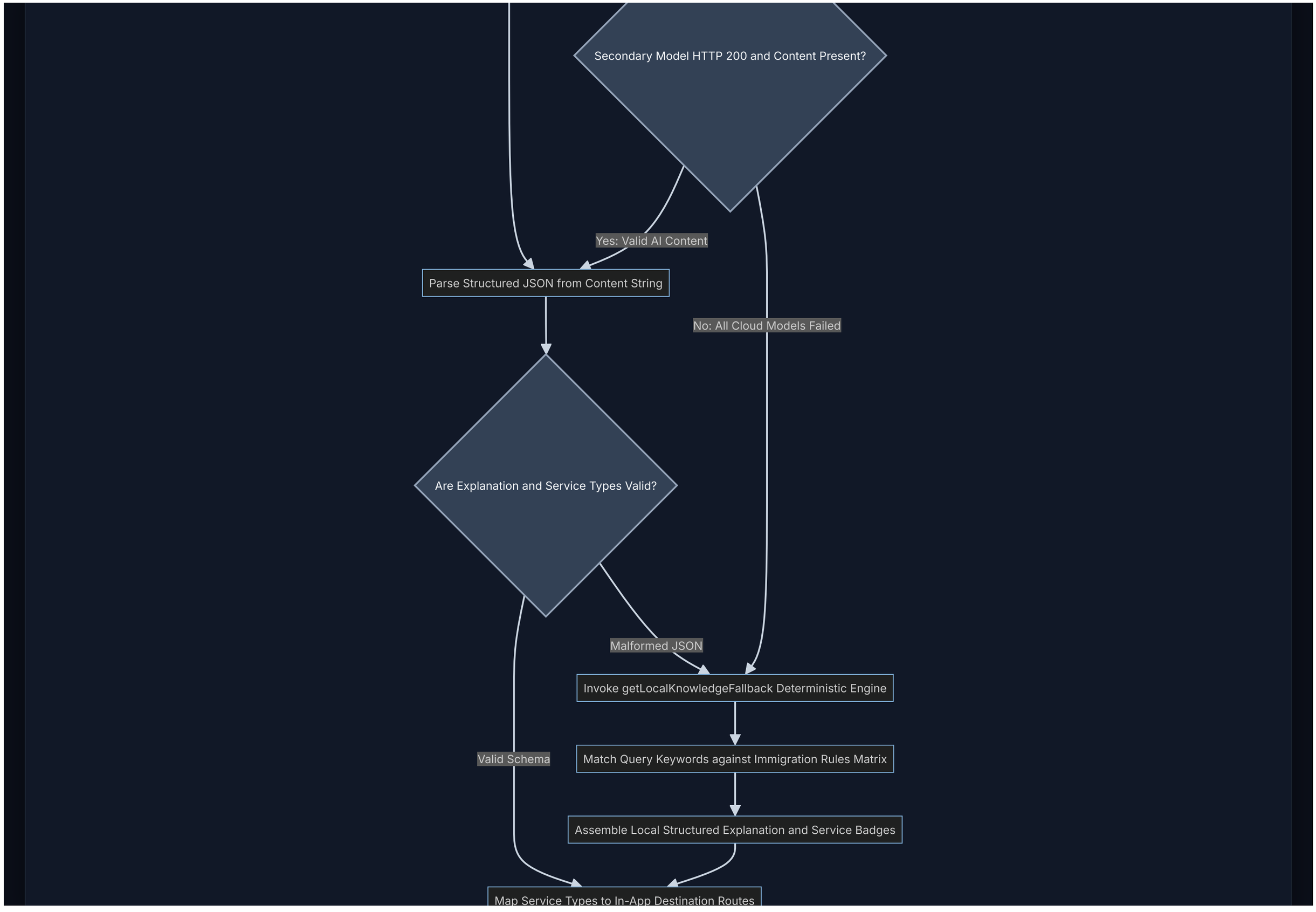

Assemble Project-OSRM Universal Endpoint with Coordinates
Execute HTTP GET to OSRM with AbortSignal Timeout 6000ms
OSRM Returned Route Geometry?
Yes: Route Present
Yes: Route Present
No: Outage or Network Error
Extract Route Geometry Coordinates Array
Extract OSRM Geometry Coordinates Array
Execute Direct Haversine Distance Fallback
Invert Coordinate Elements from Longitude-Latitude to Latitude-Longitude
Generate Direct Linear Waypoints from Origin to Destination
Calculate Estimated Travel Duration at Average Speed 40 kmh
Package Result Object with Coordinates, Distance Meters, Duration Seconds
Return Standardized Route

MICRO-PROCESS LOGIC

DFD Level 3.3: AI Conversational Reasoning Pipeline

Prompt preparation, 4-turn dialog history slicing, primary/fallback cloud LLM execution, offline keyword regex matching, and navigation chip generation.

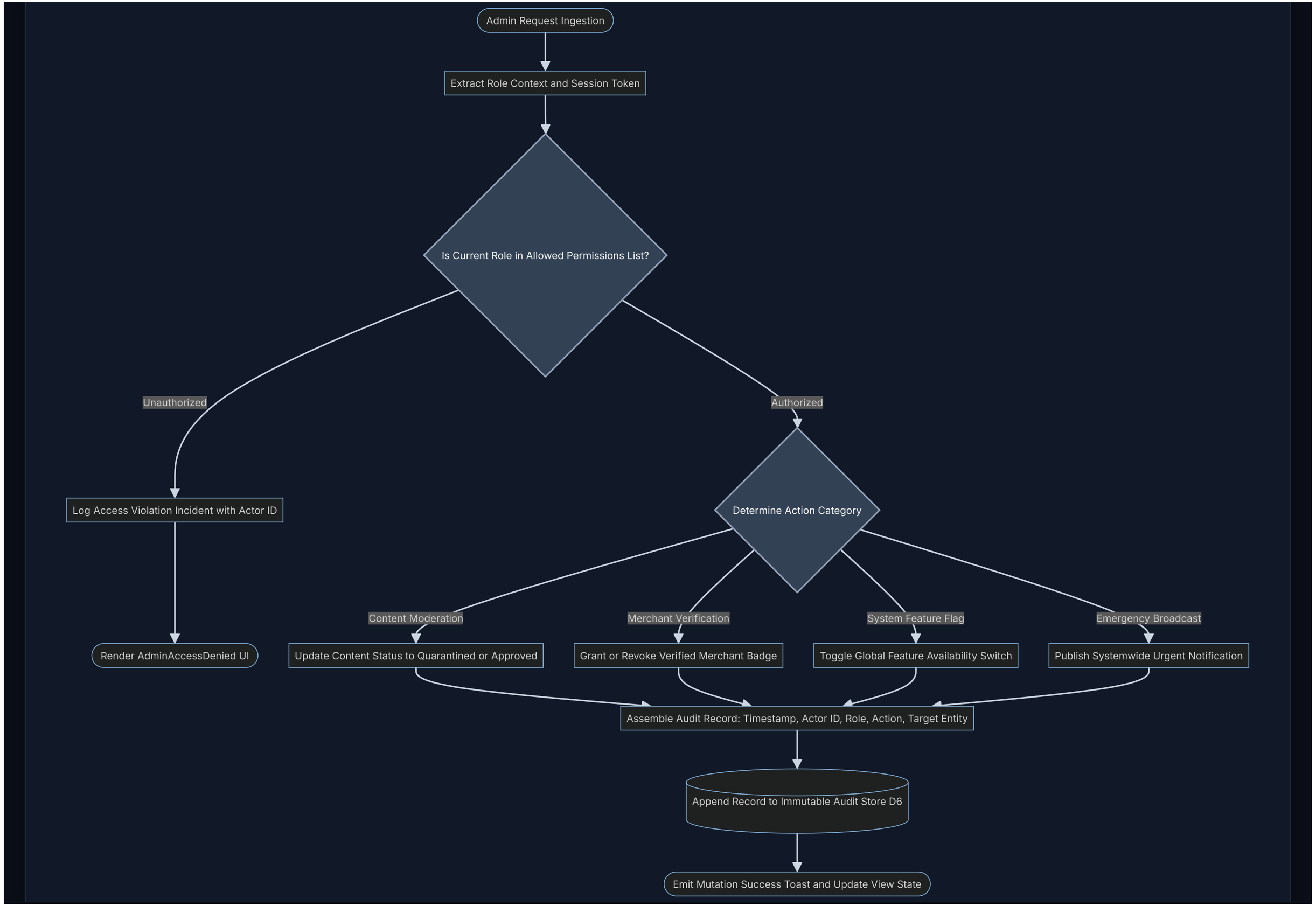

User Enters Assistant Prompt
Trim Query and Detect Target Language
Slice Last 4 Messages from Chat History
Construct Message Array: System Prompt, Dialog History, User Query
Send POST to OpenRouter Primary LLM (Mistral/Claude)
Primary Model HTTP 200 and Content Present?
No: Timeout or 4xx/5xx
Send POST to Fallback Model (Gemini Flash)
Yes: Valid AI Content


Secondary Model HTTP 200 and Content Present?
Yes: Valid AI Content
Parse Structured JSON from Content String
No: All Cloud Models Failed
Are Explanation and Service Types Valid?
Malformed JSON
Invoke getLocalKnowledgeFallback Deterministic Engine
Valid Schema
Match Query Keywords against Immigration Rules Matrix
Assemble Local Structured Explanation and Service Badges
Map Service Types to In-App Destination Routes

MICRO-PROCESS LOGIC

DFD Level 3.4: Admin RBAC Guard & Audit Log Pipeline

Role credential extraction, permission authorization check, administrative mutation execution (moderation, verification, flags, alerts), and immutable audit record appending.


Admin Request Ingestion
Extract Role Context and Session Token
Is Current Role in Allowed Permissions List?
Unauthorized
Authorized
Log Access Violation Incident with Actor ID
Render AdminAccessDenied UI
Determine Action Category
Content Moderation
Merchant Verification
System Feature Flag
Emergency Broadcast
Update Content Status to Quarantined or Approved
Grant or Revoke Verified Merchant Badge
Toggle Global Feature Availability Switch
Publish Systemwide Urgent Notification
Assemble Audit Record: Timestamp, Actor ID, Role, Action, Target Entity
Append Record to Immutable Audit Store D6
Emit Mutation Success Toast and Update View State

DATA ARCHITECTURE

Data Dictionary & Data Stores Schema

Complete structural schema, storage mechanisms, primary keys, and retention lifecycles for Data Stores D1 through D7.

STORE ID	STORE NAME	STORAGE MECHANISM	SCHEMA & KEY ATTRIBUTES	RETENTION & SCOPE
`D1`	**Session & Tab Memory Store**	`localStorage` / `MobileTabContext`	`{ activeTab: string, tabHistory: string[], tabs: [{ id, title, path, snapshot }] }`	Persistent across app restarts; device-bound
`D2.1`	**User Coordinates Cache**	`localStorage["bkoi_last_user_coords"]`	`{ lat: number, lng: number, timestamp: number }`	24 Hours validity; eliminates redundant GPS prompts
`D2.2`	**Geocoding Address Cache**	In-Memory `Map<string, BariKoiAddressInfo>`	`Key: "lat.3,lng.3" → { address, area, district, postCode, city, sub_district }`	Ephemeral runtime; instant sub-millisecond reverse lookups
`D2.3`	**BariKoi Circuit Breaker**	`sessionStorage["bkoi_cooldown_until"]`	`Timestamp (ms)`	30 Seconds cooldown upon HTTP 429 response
`D3`	**Curated Places & Dynamic Listings**	Static TypeScript Store & Generator	`{ id, name, category, lat, lng, rating, reviews, open, address, phone, immigrantFriendly, image }`	Immutable base dataset + real-time distance calculation
`D4`	**Community & Multimedia Store**	Component State & Local Feed	`{ id, author, avatar, timestamp, title, body, likes, comments, verifiedTag }`	Session & feed lifecycle
`D5`	**Marketplace Products & Orders**	`AccountModeContext` & Local State	`{ id, sellerId, title, price, category, stock, images, orders: [{ id, buyerId, status }] }`	Merchant & Buyer session lifecycle
`D6`	**Admin Moderation & Audit Logs**	`AdminRoleContext` & Audit Store	`{ id, actorId, role, action, targetEntity, timestamp, status: 'approved' | 'quarantined' }`	Immutable audit trail
`D7`	**AI Assistant Conversation Memory**	React State `messages[]`	`Array<{ sender: 'user' | 'bot', text: string, timestamp, serviceTypes?: [...] }>`	Last 4 conversation turns passed to OpenRouter

CODE & PROCESS TRACEABILITY

Comprehensive Function & Process Mapping Matrix

Direct mapping linking system processes to concrete React components, services, exported functions, and dataflow signatures in the repository.

PROCESS ID	PROCESS NAME	IMPLEMENTATION FILE	CORE FUNCTION / SYMBOL	DATA INPUT ➔ OUTPUT FLOW
`1.1`	Tab Management & Navigation	`MobileNav.tsx`	`MobileNav()`	User click ➔ Active tab route switch & scroll event listener
`1.2`	Recent Tabs Switcher	`MobileMultiTabSwitcher.tsx`	`MobileMultiTabSwitcher()`	Active tabs array ➔ Android-style recent cards view
`1.3`	Multi-Country Management	`CountryPlatformContext.tsx`	`CountryPlatformProvider()`	Country selection ➔ Dynamic bounding box, flag & currency tokens
`1.4`	Buyer vs Seller SaaS Mode	`AccountModeContext.tsx`	`useAccountMode()`	Toggle switch ➔ Consumer view vs Merchant Storefront Dashboard
`2.1`	Reverse Geocode with Caching	`barikoiService.ts`	`safeBariKoiReverseGeocode()`	`(lat, lng, countryCode)` ➔ Cached/Fetched `BariKoiAddressInfo`
`2.2`	429 Cooldown Trip	`barikoiService.ts`	`triggerBariKoiCooldown()`	HTTP 429 received ➔ 30,000ms cooldown window in `sessionStorage`
`2.3`	BariKoi GL WebGL Loader	`mapboxService.ts`	`loadBkoiGL()`	CDN script tag injection ➔ Patched `window.bkoigl.Map` constructor
`2.4`	502/CORS Error Interceptor	`mapboxService.ts`	`attachMapboxFallbackOnError()`	WebGL tile error event ➔ Seamless switch to Mapbox/OSM raster style
`2.5`	Dynamic Watermark Badge	`mapboxService.ts`	`setupMapWatermark()`	Provider change ➔ BariKoi / Mapbox / OSM SVG logo badge overlay
`3.1`	Generic Service Directory Map	`ServiceMapDirectory.tsx`	`ServiceMapDirectory()`	21 Sub-services ➔ Interactive WebGL map, bottom sheet & cards
`3.2`	Haversine Distance Calculation	`ServiceMapDirectory.tsx`	`calculateDistanceKm()`	`(lat1, lon1, lat2, lon2)` ➔ Distance in kilometers
`3.3`	Camera Pan Snapping Guard	`ServiceMapDirectory.tsx`	`lastFocusedListingIdRef`	User drag/pan ➔ Suppresses automatic camera re-centering
`4.1`	Turn-by-Turn Global Routing	`mapboxService.ts`	`fetchGlobalRoute()`	`(from, to, mode)` ➔ GeoJSON coordinates, distance meters, duration seconds
`5.1`	Housing Purpose & Rent Filter	`Housing.tsx`	`togglePurpose()`	"Rent" vs "Buy" click ➔ Filtered real-estate listings

PROCESS ID	PROCESS NAME	IMPLEMENTATION FILE	CORE FUNCTION / SYMBOL	DATA INPUT ➔ OUTPUT FLOW
`5.2`	Dynamic Jobs Ingestion	`Jobs.tsx`	`JobsPage()`	User location ➔ Localized dynamic warehouse, driver, retail jobs
`7.1`	OpenRouter AI Agent Invocator	`openRouterService.ts`	`callOpenRouterAgent()`	`(query, lang, history)` ➔ Structured reasoning & service navigation tokens
`7.2`	Offline Immigrant FAQ Fallback	`openRouterService.ts`	`getLocalKnowledgeFallback()`	Query regex ➔ Instant offline answer & category route suggestions
`8.1`	6-Tier RBAC Route Guard	`AdminRoleGuard.tsx`	`AdminRoleGuard()`	Current role vs Allowed roles ➔ Route render or `AdminAccessDenied`
`8.2`	Super Admin Feature Flags	`AdminFeatureFlags.tsx`	`AdminFeatureFlags()`	Switch toggle ➔ Instant global enable/disable of platform services

